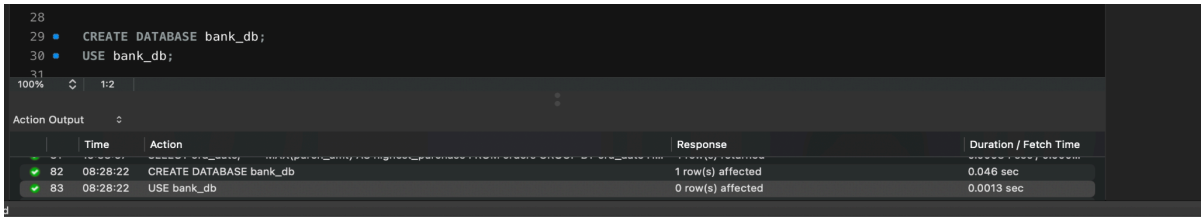
SQL Practice — Bank Management System

AGGREGATE FUNCTIONS with WHERE, GROUP BY, HAVING, ORDER BY

Name: Hansika
Roll No: 2520030237
Section: 7
Email: gunapatihansikareddy@klh.edu.in

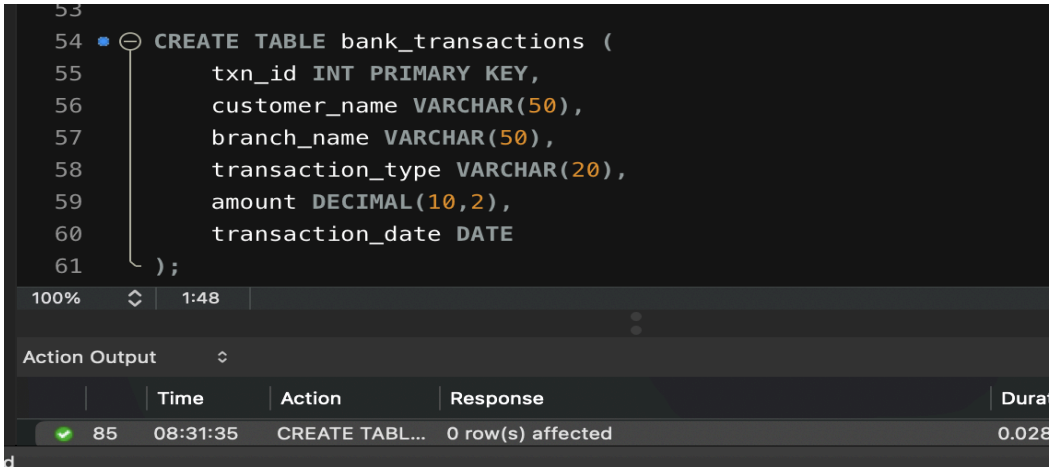
Step 1 — Create Database

```
CREATE DATABASE bank_db;
USE bank_db;
```



Step 2 — Create Bank_Transactions Table (DDL)

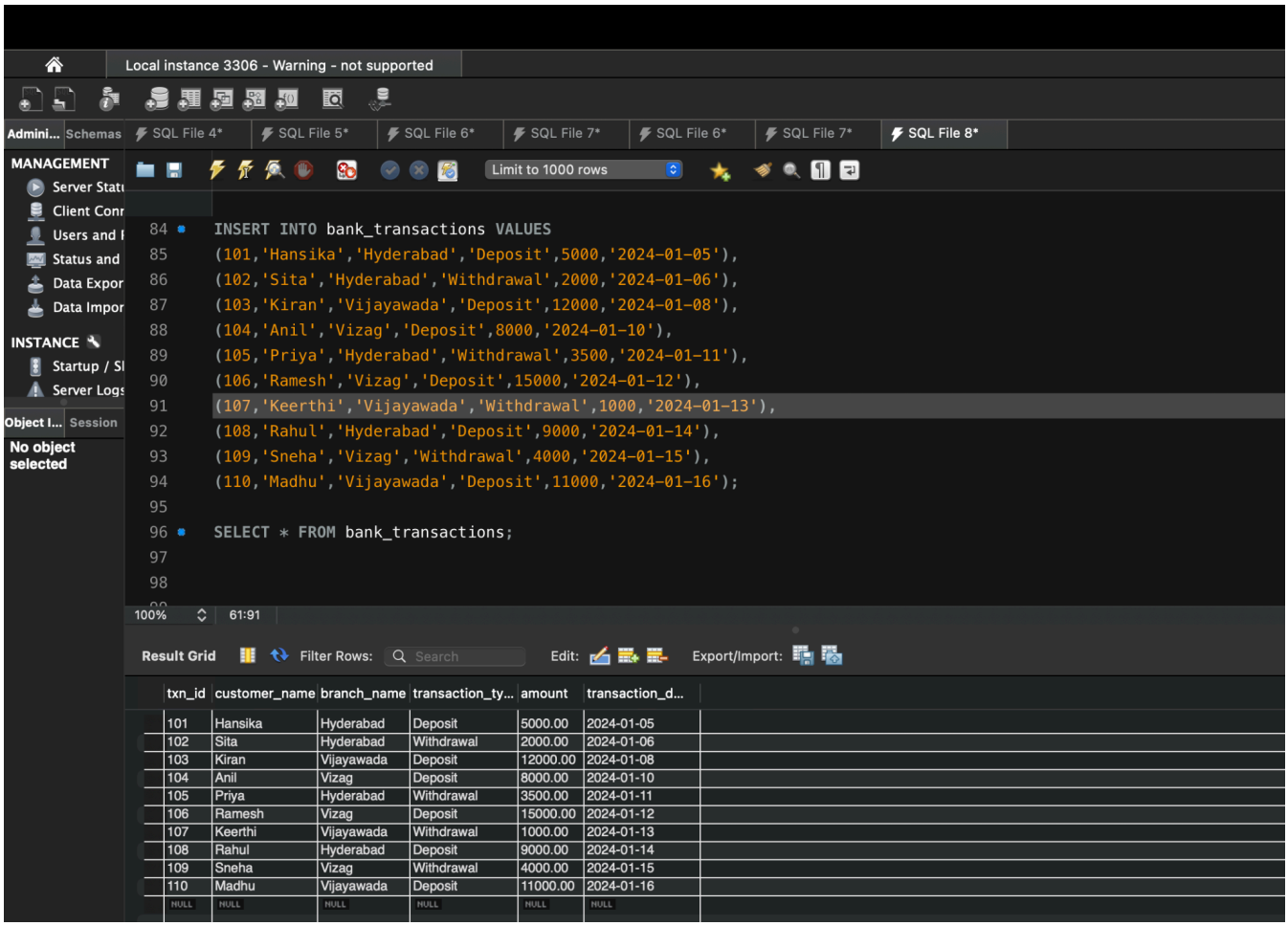
```
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```



Step 3 — Insert Data and Display

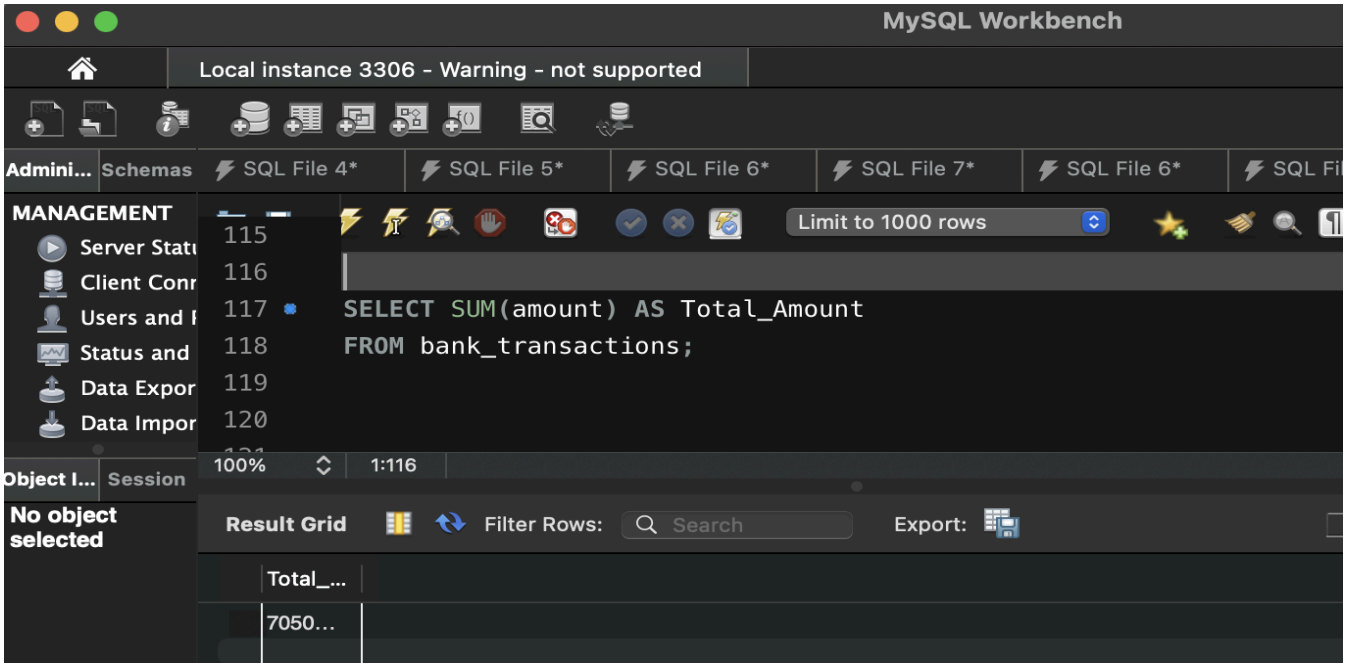
```
INSERT INTO bank_transactions VALUES
(101,'Hansika','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
(105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
(107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');

SELECT * FROM bank_transactions;
```



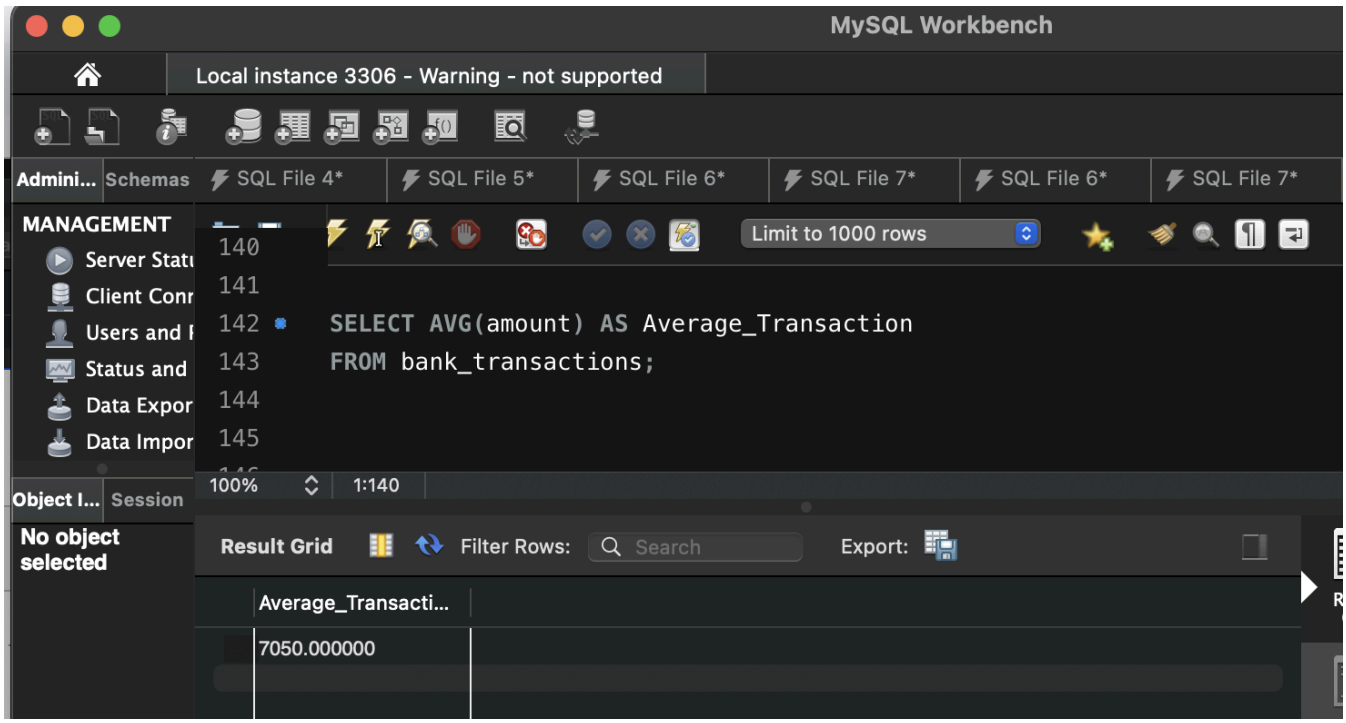
Step 4 — Question 1: SUM()

```
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;
```



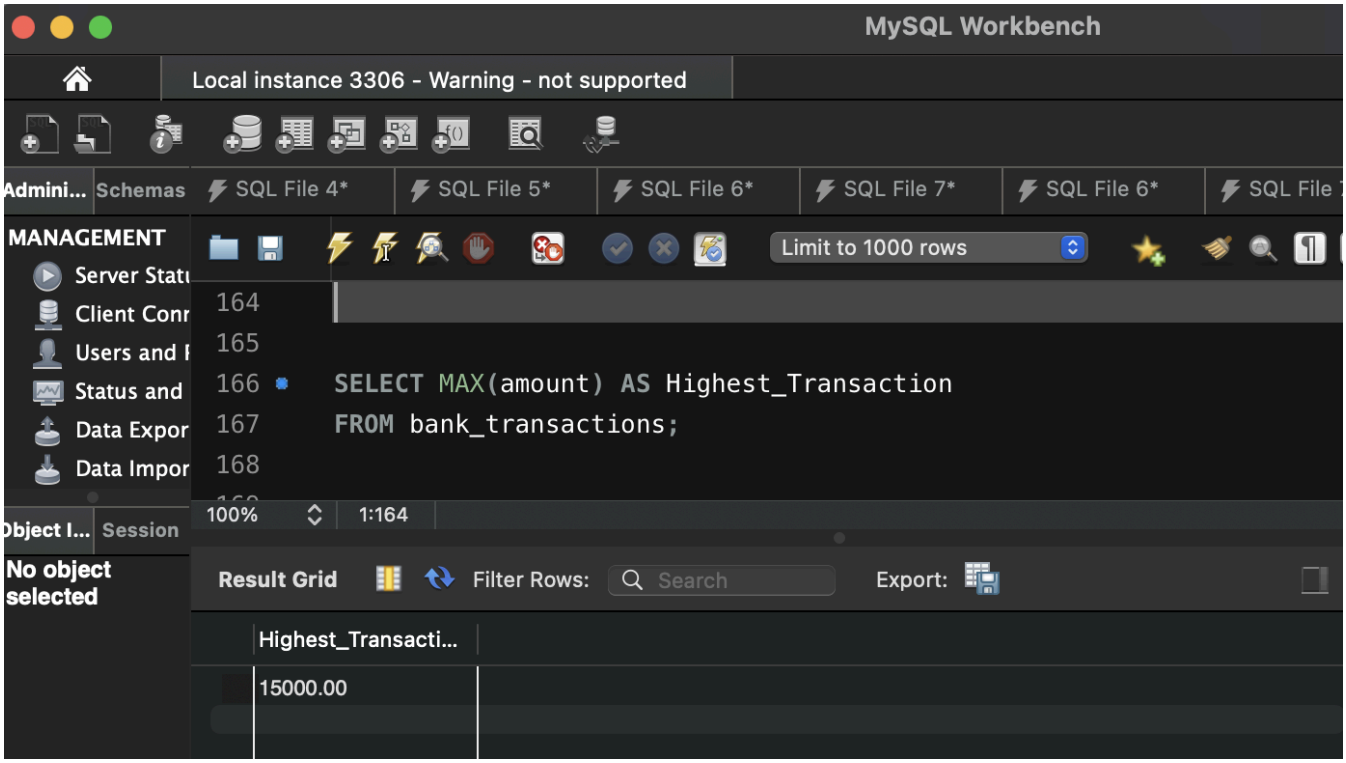
Step 5 — Question 2: AVG()

```
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;
```



Step 6 — Question 3: MAX()

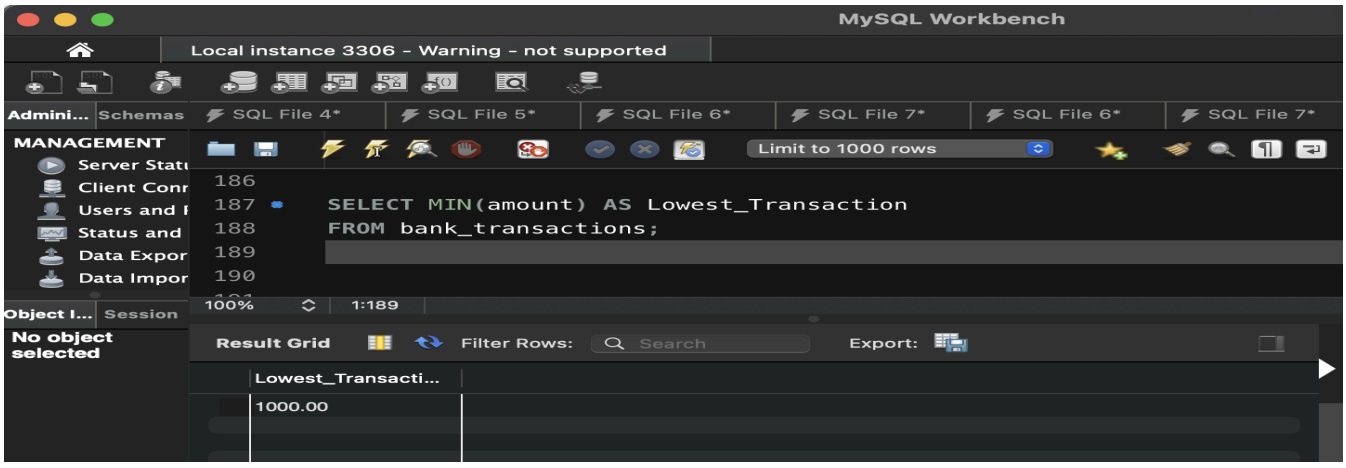
```
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;
```



Step 7 — Question 4: MIN()

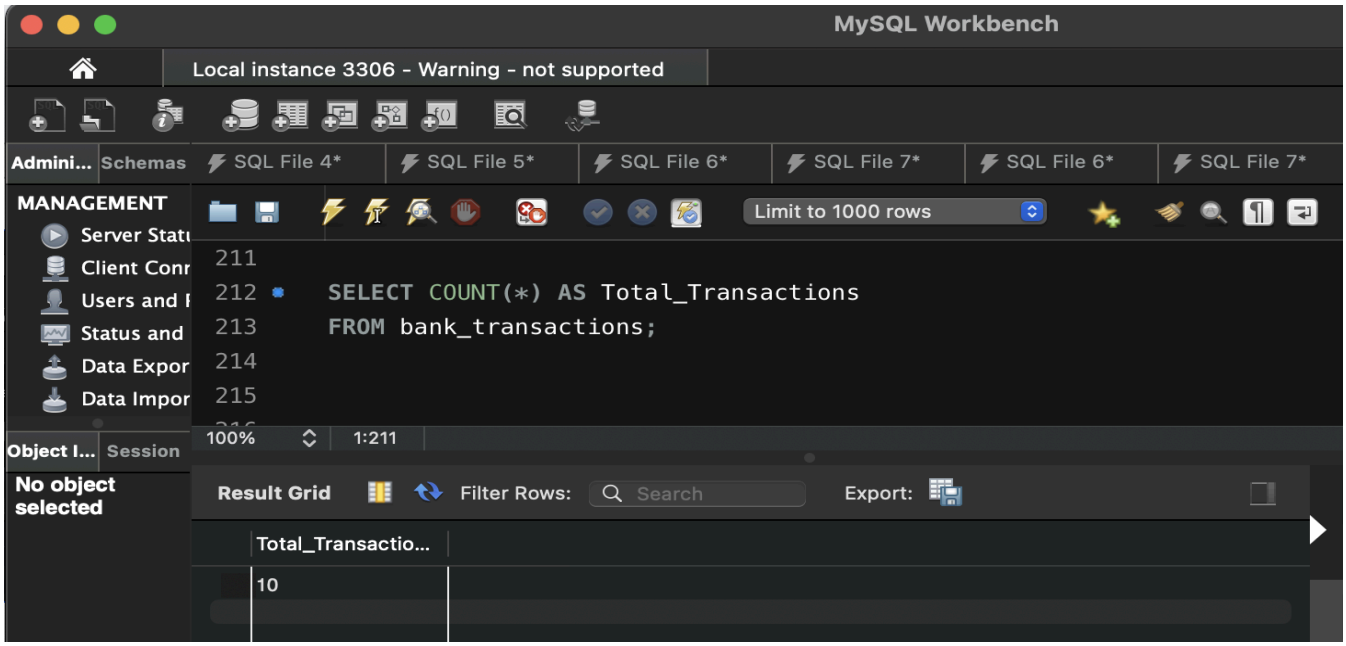
Scenario: The audit department wants to know the smallest transaction amount recorded.

```
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```



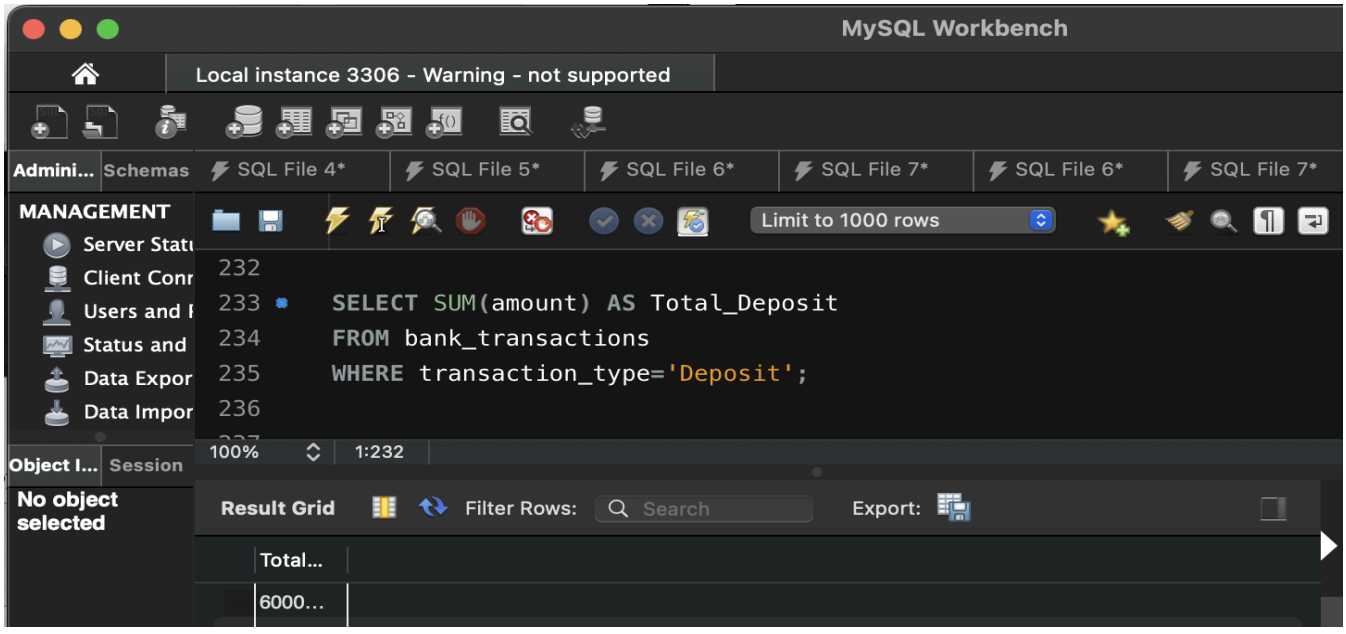
Step 8 — Question 5: COUNT()

```
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```



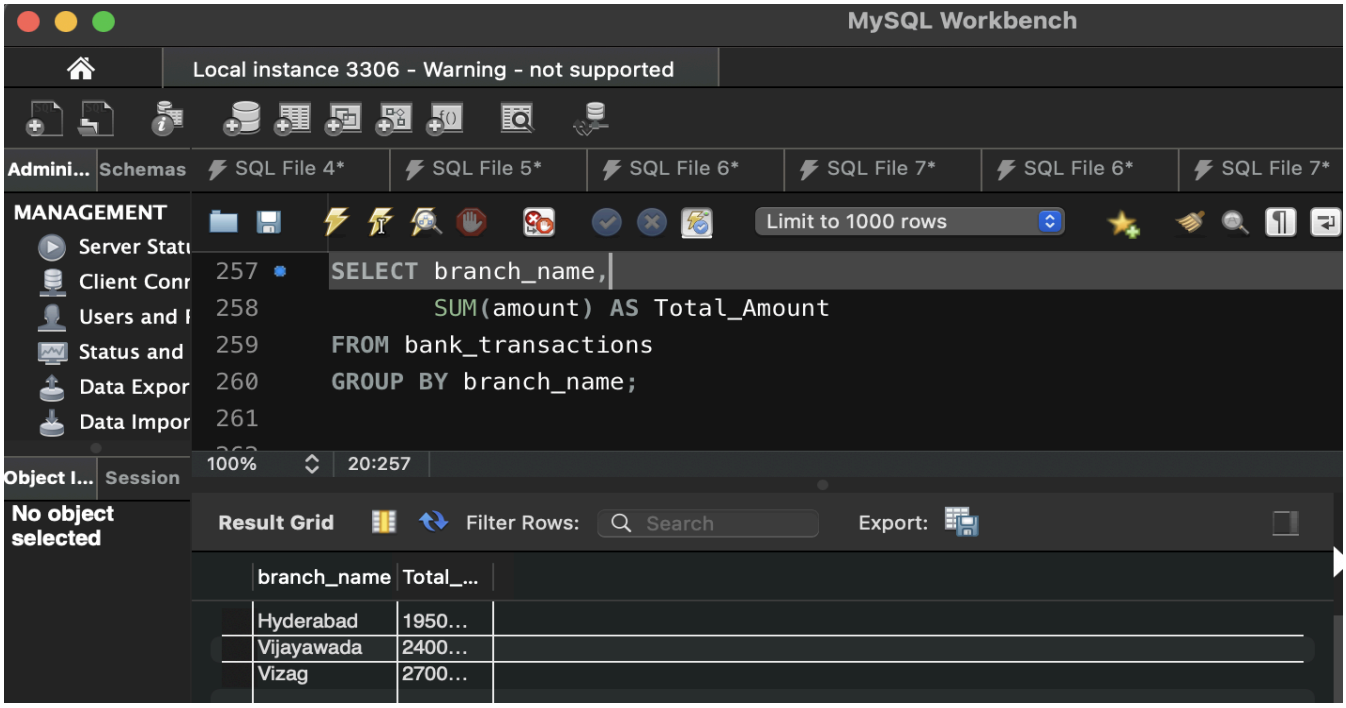
Step 9 — Question 6: WHERE + SUM()

```
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```



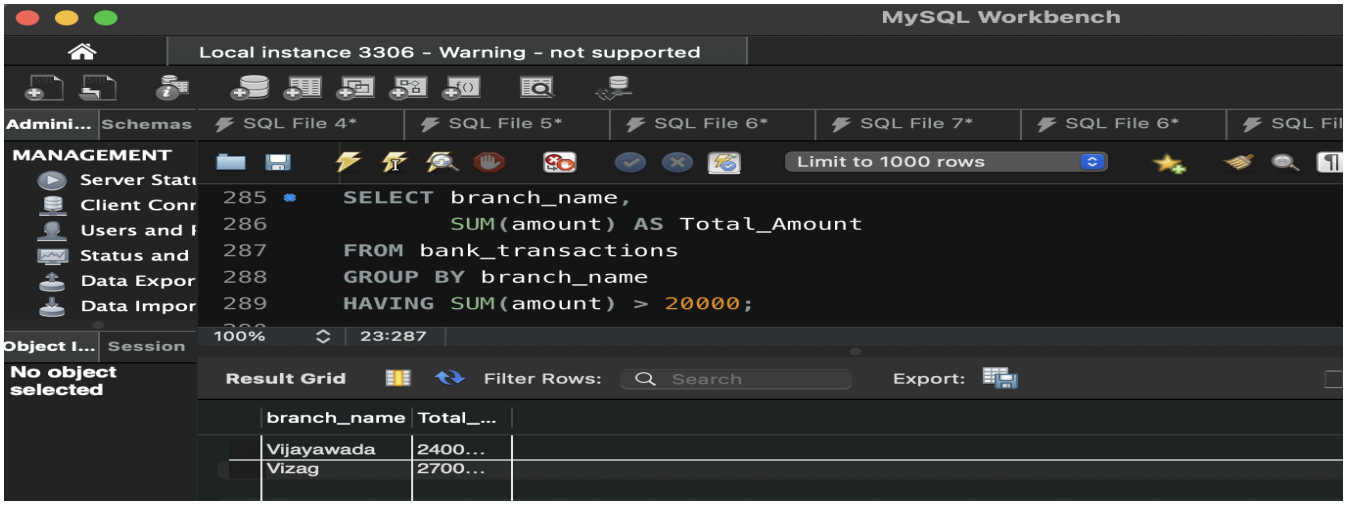
Step 10 — Question 7: GROUP BY

```
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```



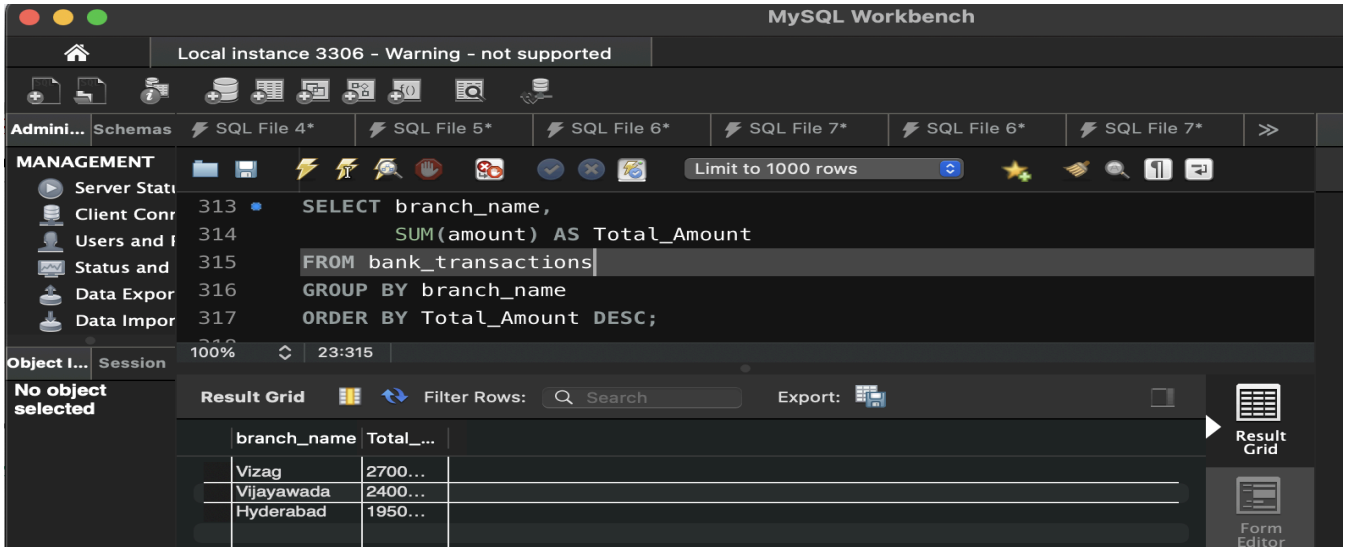
Step 11 — Question 8: GROUP BY + HAVING

```
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;
```



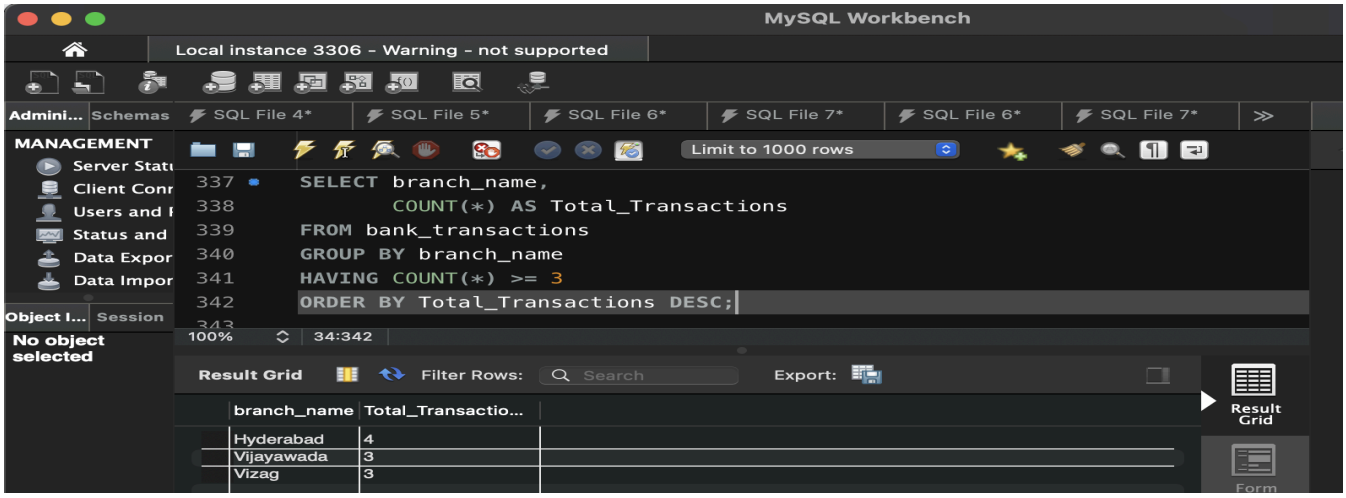
Step 12 — Question 9: GROUP BY + ORDER BY

```
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```



Step 13 — Question 10: GROUP BY + HAVING + ORDER BY

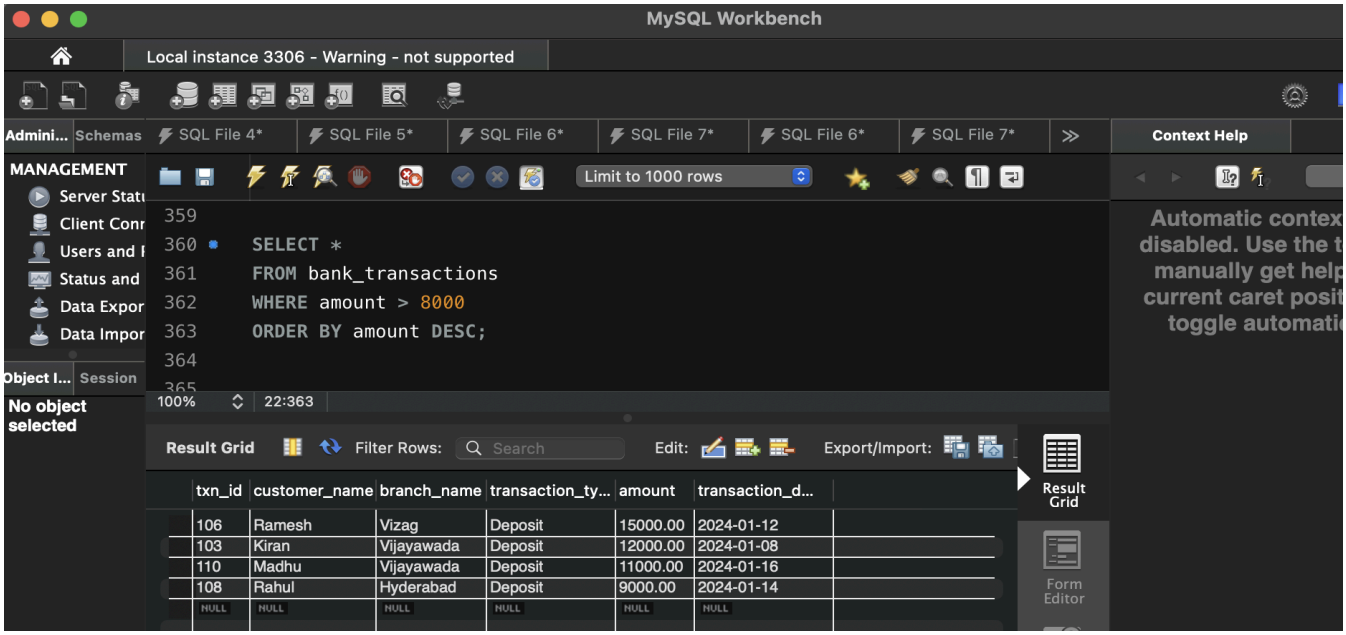
```
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```



Step 14 — Question 11: WHERE + ORDER BY

Scenario: The fraud detection team wants all transactions greater than $8,000.

```
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```



Step 15 — Question 12: WHERE + GROUP BY + HAVING

```
SELECT branch_name,
       SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```